

Using Encryption to Enhance Confidentiality and Integrity (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 05

Student:
James Collins

Email:
collins222@usf.edu

Time on Task:
1 hour, 36 minutes

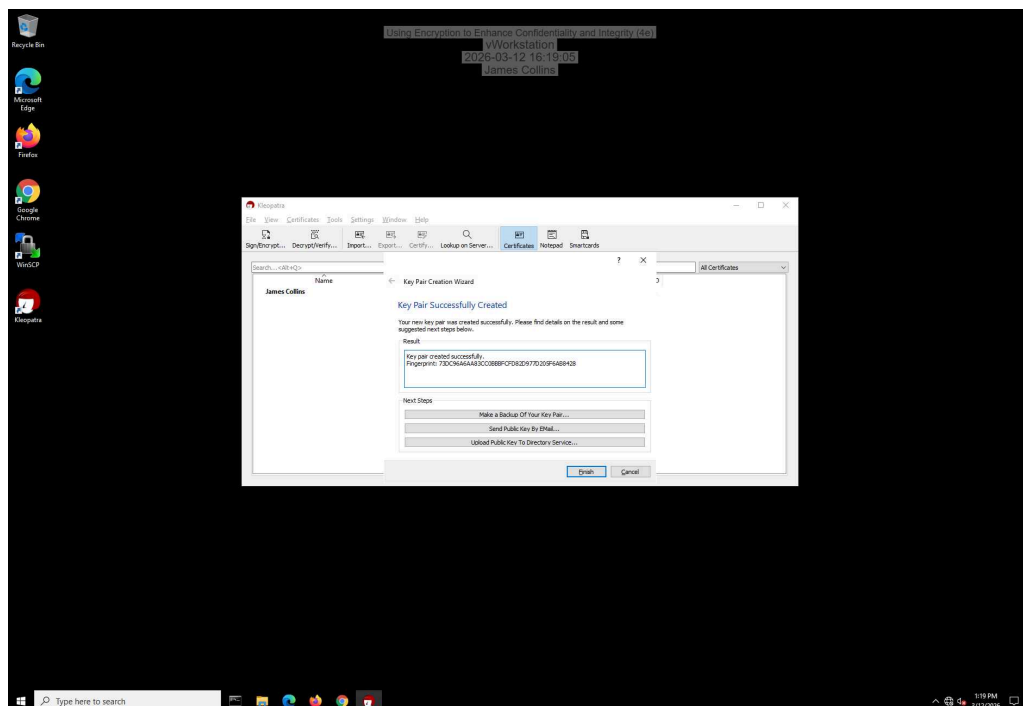
Progress:
100%

Report Generated: Sunday, March 15, 2026 at 2:32 PM

Section 1: Hands-On Demonstration

Part 1: Create and Exchange Asymmetric Encryption Keys

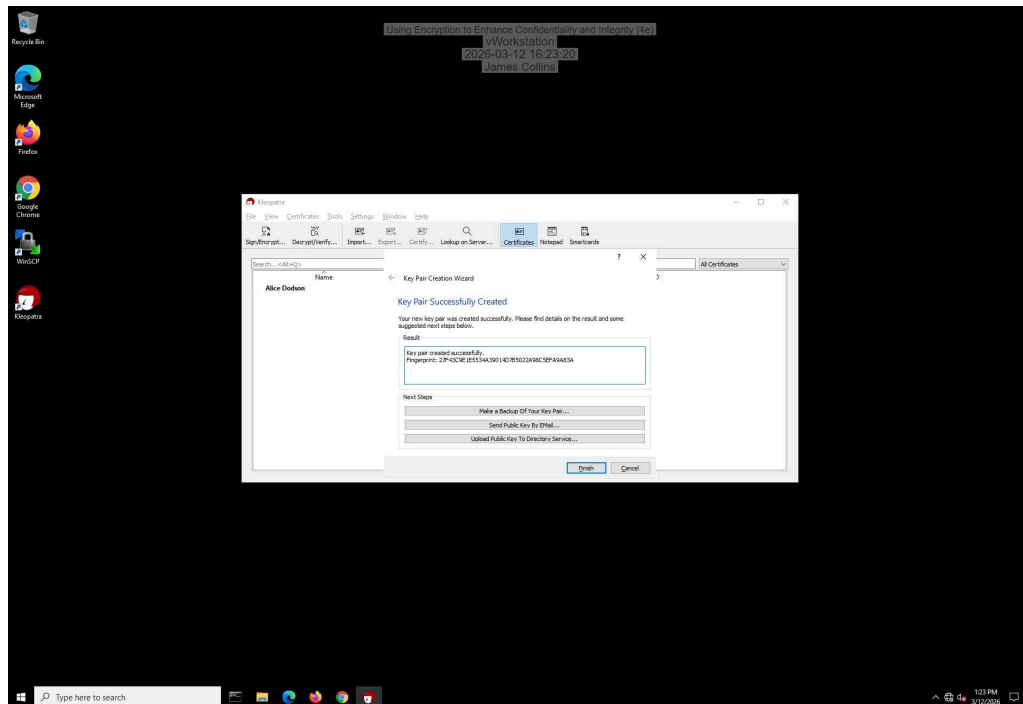
9. Make a screen capture showing the **fingerprint** for your key pair.



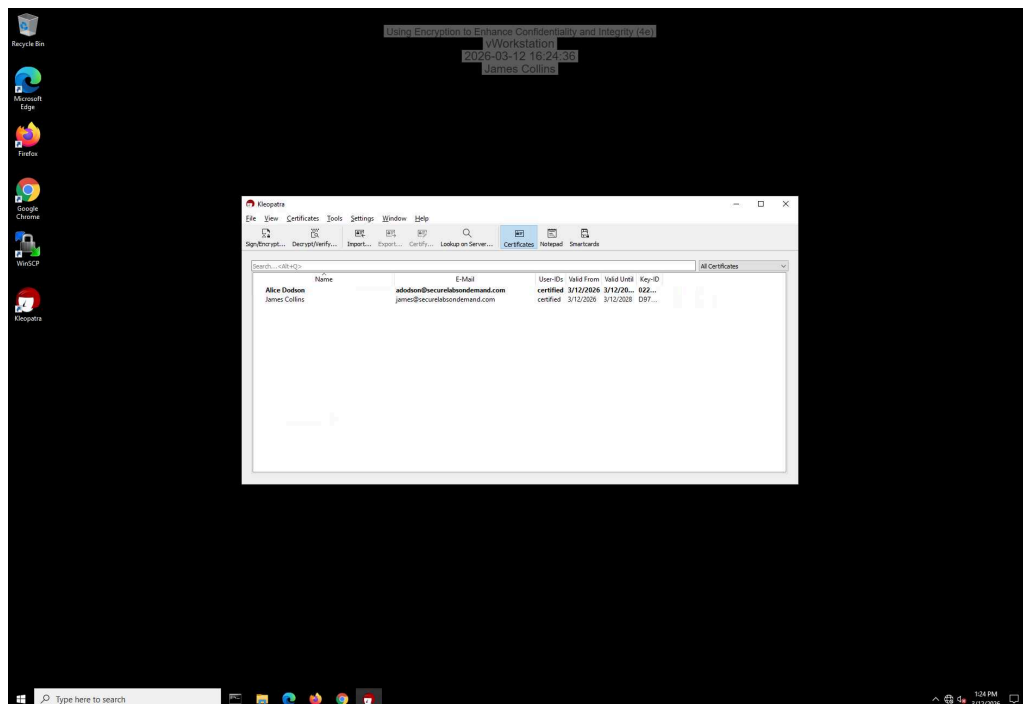
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22. Make a screen capture showing the **fingerprint** for Alice's key pair.



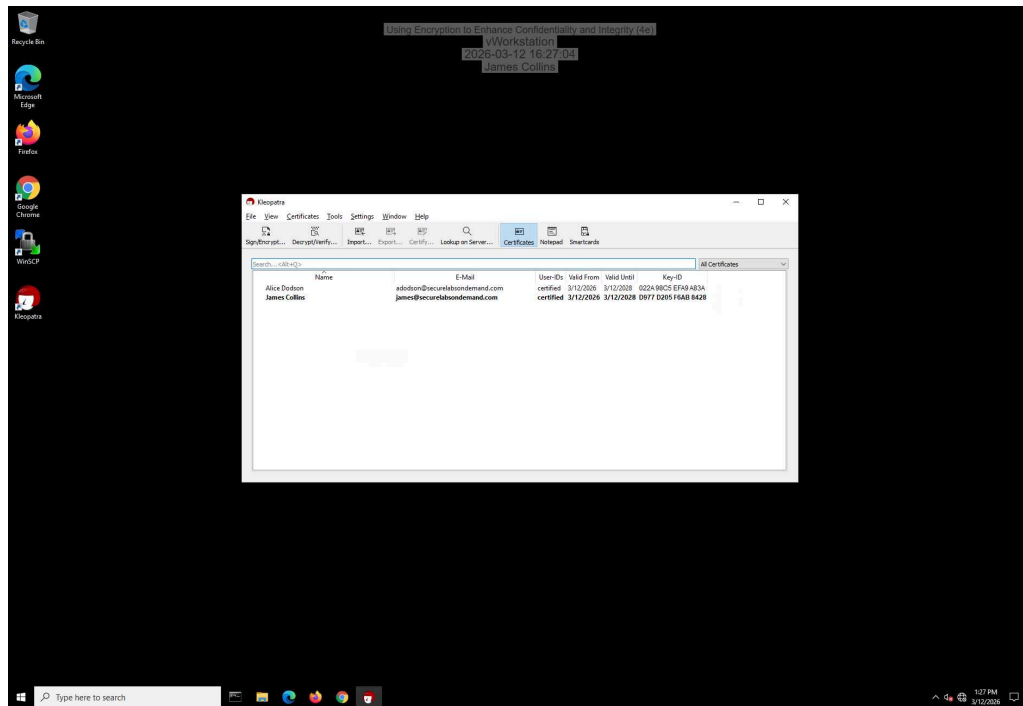
30. Make a screen capture showing your **public key** in Alice's certificate cache.



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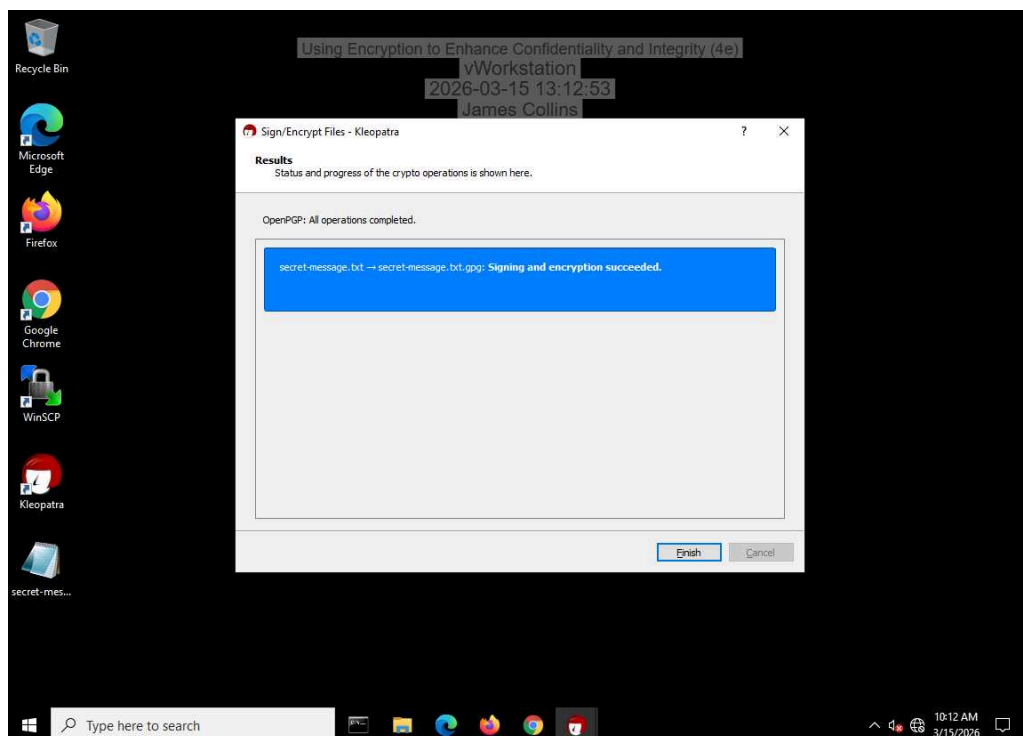
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35. Make a screen capture showing Alice's public key in your certificate cache.

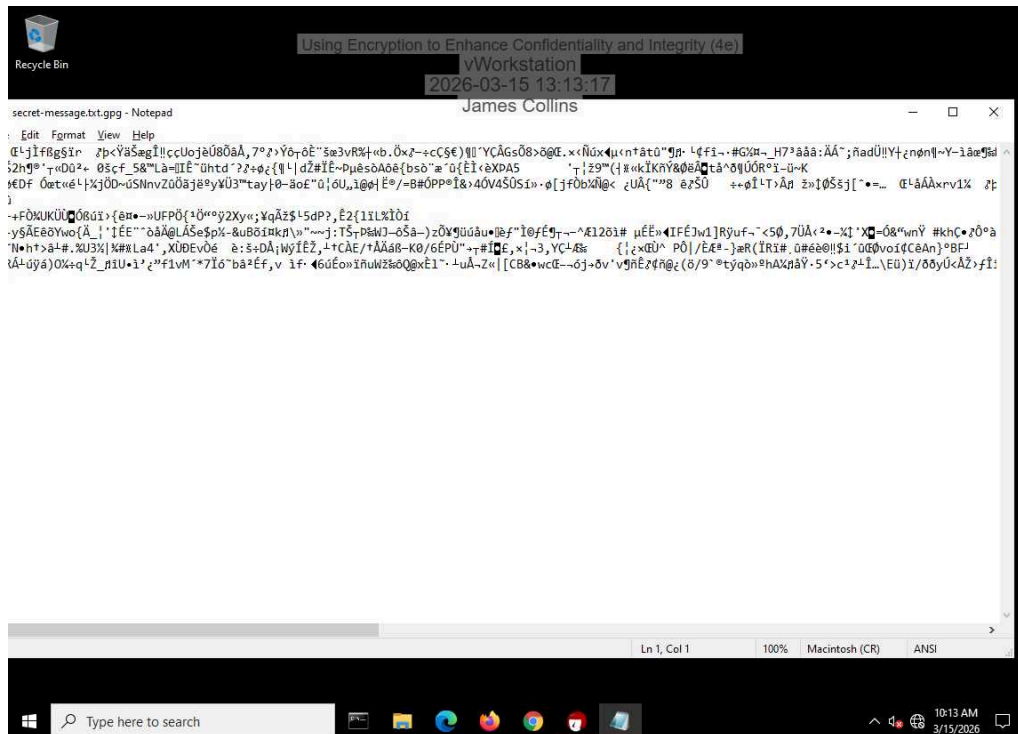


Part 2: Encrypt a File Using Asymmetric Encryption

9. Make a screen capture showing the successful signing and encryption message.



12. Make a screen capture showing the ciphertext.

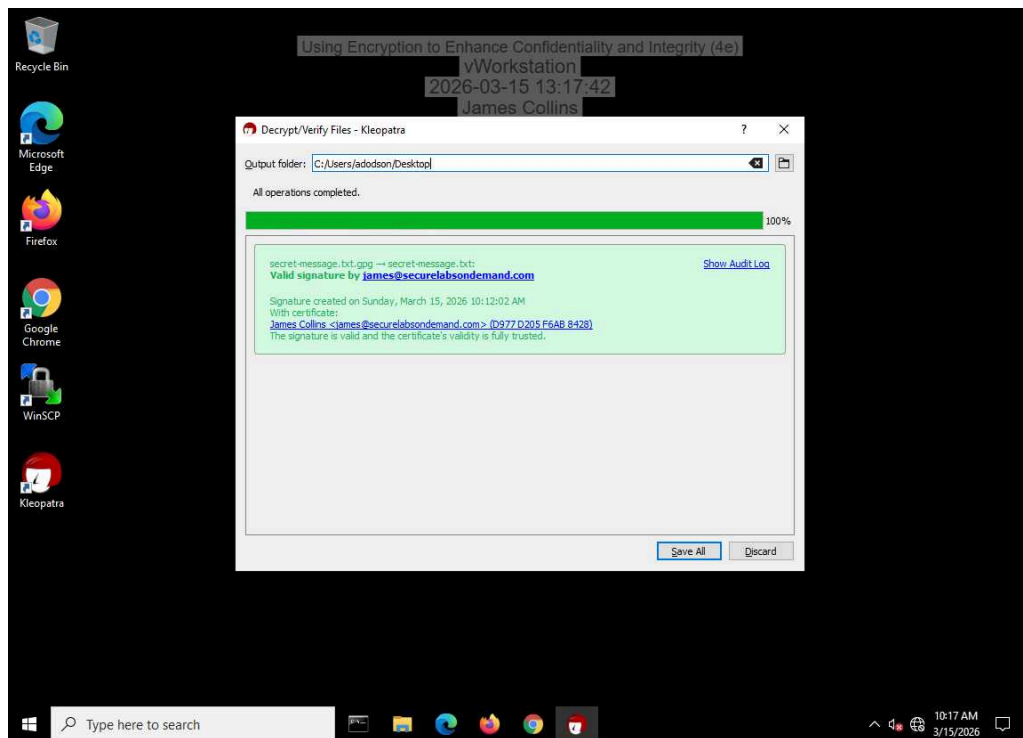


Part 3: Decrypt a File Using Asymmetric Encryption

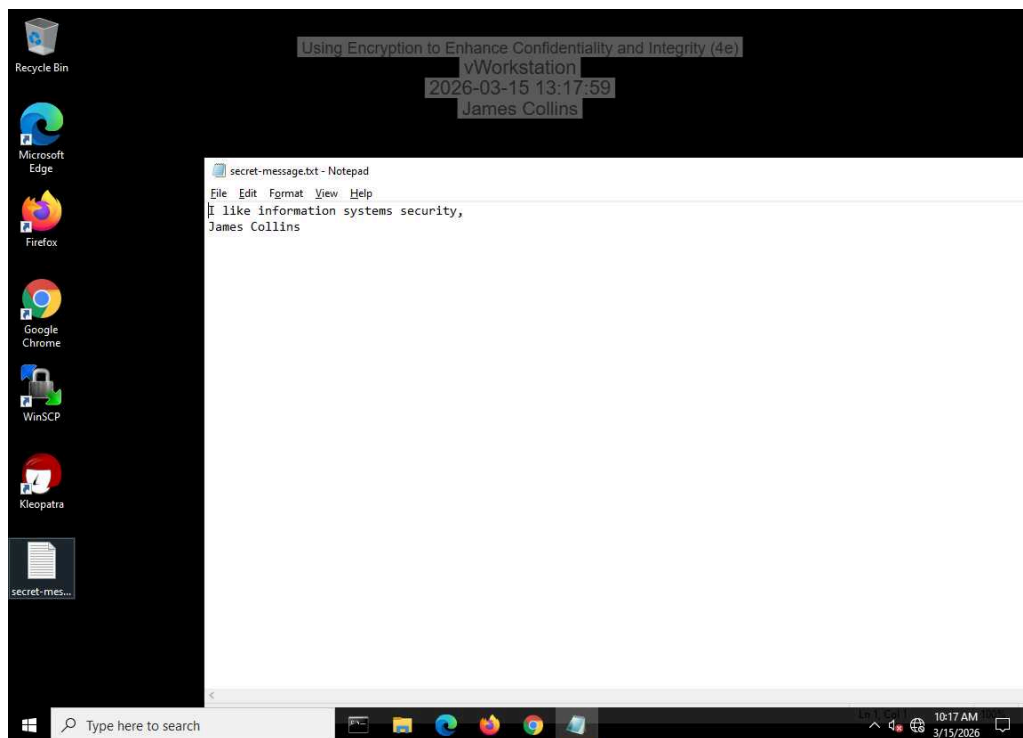
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15. Make a screen capture showing the **Decrypt/Verify Files** window.



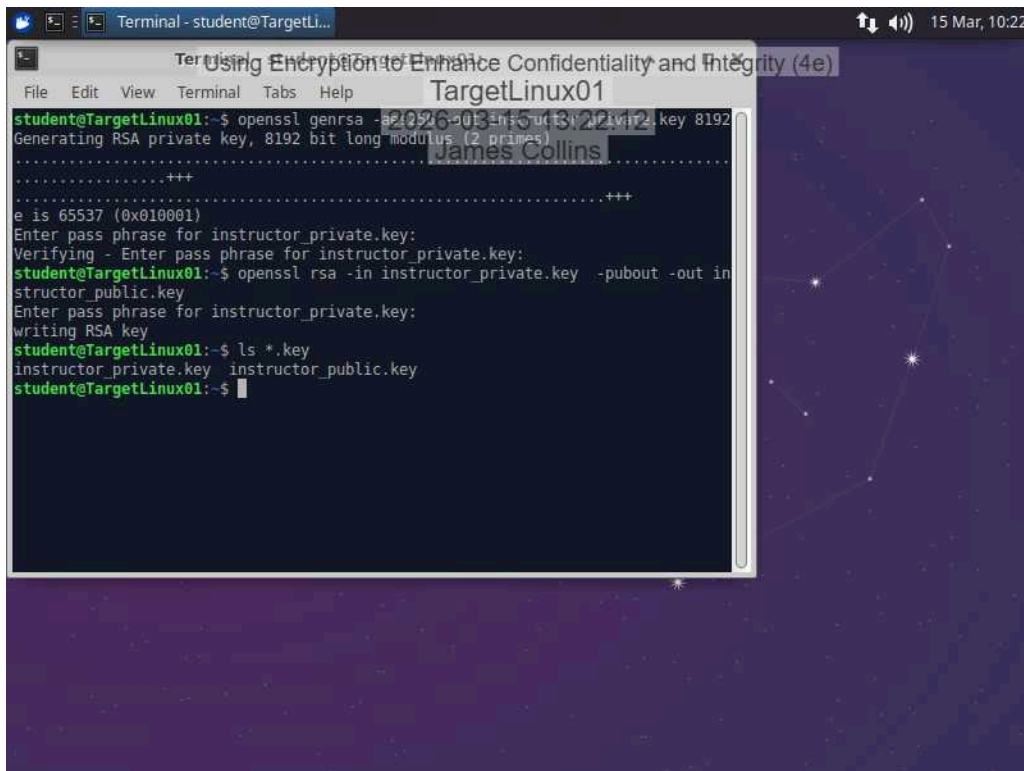
18. Make a screen capture showing the **decrypted secret-message.txt** file in Notepad.



Section 2: Applied Learning

Part 1: Create an Asymmetric Key Pair

10. Make a screen capture showing the instructor's key pair files.



The screenshot shows a terminal window titled "Terminal - student@TargetLinux01" with a menu bar (File, Edit, View, Terminal, Tabs, Help). The terminal output is as follows:

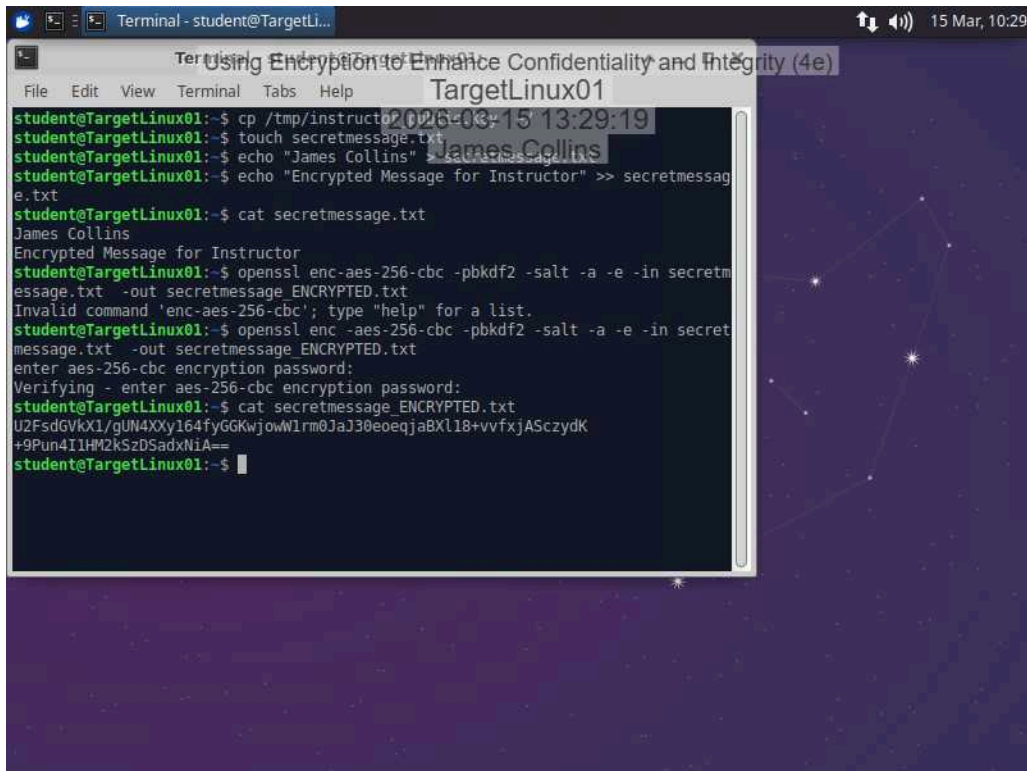
```
student@TargetLinux01:~$ openssl genrsa -out instructor_private.key 8192
Generating RSA private key, 8192 bit long modulus (2 primes)
.....+++
e is 65537 (0x010001)
Enter pass phrase for instructor_private.key:
Verifying - Enter pass phrase for instructor_private.key:
student@TargetLinux01:~$ openssl rsa -in instructor_private.key -pubout -out instructor_public.key
Enter pass phrase for instructor_private.key:
writing RSA key
student@TargetLinux01:~$ ls *.key
instructor_private.key  instructor_public.key
student@TargetLinux01:~$
```

Part 2: Encrypt a File Using Symmetric Encryption

11. Document the password you used to symmetrically encrypt the file.

password

13. Make a screen capture showing the ciphertext in the `secretmessage_ENCRYPTED.txt` file.



```
student@TargetLinux01:~$ cp /tmp/instructor/026-03-15 13:29:19
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo "James Collins" > secretmessage.txt
student@TargetLinux01:~$ echo "Encrypted Message for Instructor" >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
James Collins
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc-aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
Invalid command 'enc-aes-256-cbc'; type "help" for a list.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2Fs dGVkX1/gUN4XXyl64fyGGKwjowW1rm0JaJ30eoeqjaBXl18+vvfxjASczydK
+9Pun4I1HM2KSzDSadxNiA==
student@TargetLinux01:~$
```

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16. Make a screen capture showing the output of the ls command.

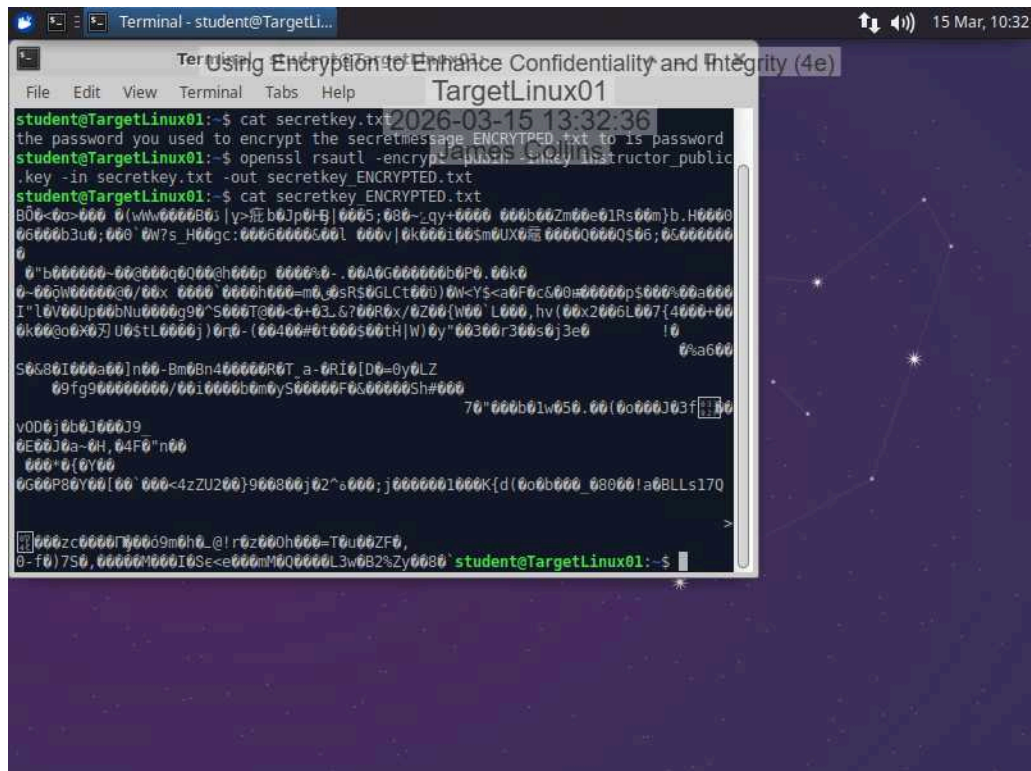
```
student@TargetLinux01:~$ cp /tmp/instructor_private.key .
student@TargetLinux01:~$ touch secretmessage.txt
student@TargetLinux01:~$ echo "James Collins" > secretmessage.txt
student@TargetLinux01:~$ echo "Encrypted Message for Instructor" >> secretmessage.txt
student@TargetLinux01:~$ cat secretmessage.txt
James Collins
Encrypted Message for Instructor
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
Invalid command 'enc-aes-256-cbc'; type 'help' for a list.
student@TargetLinux01:~$ openssl enc -aes-256-cbc -pbkdf2 -salt -a -e -in secretmessage.txt -out secretmessage_ENCRYPTED.txt
enter aes-256-cbc encryption password:
Verifying - enter aes-256-cbc encryption password:
student@TargetLinux01:~$ cat secretmessage_ENCRYPTED.txt
U2FsdGVkX1/gUN4XXy164fyGGKwjowW1rm0JaJ30eoeqja8Xl18+vvfxjASczydK
+9Pun4I1HM2kSzDSadxNIA==
student@TargetLinux01:~$ rm secretmessage.txt
student@TargetLinux01:~$ ls
Desktop  instructor_private.key  Pictures  Templates
Documents  instructor_public.key  Public    Videos
Downloads  Music                  secretmessage_ENCRYPTED.txt
```

Part 3: Transfer and Decrypt a File Using Hybrid Cryptography

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6. Make a screen capture showing the encrypted contents of the `secretkey_ENCRYPTED.txt` file.

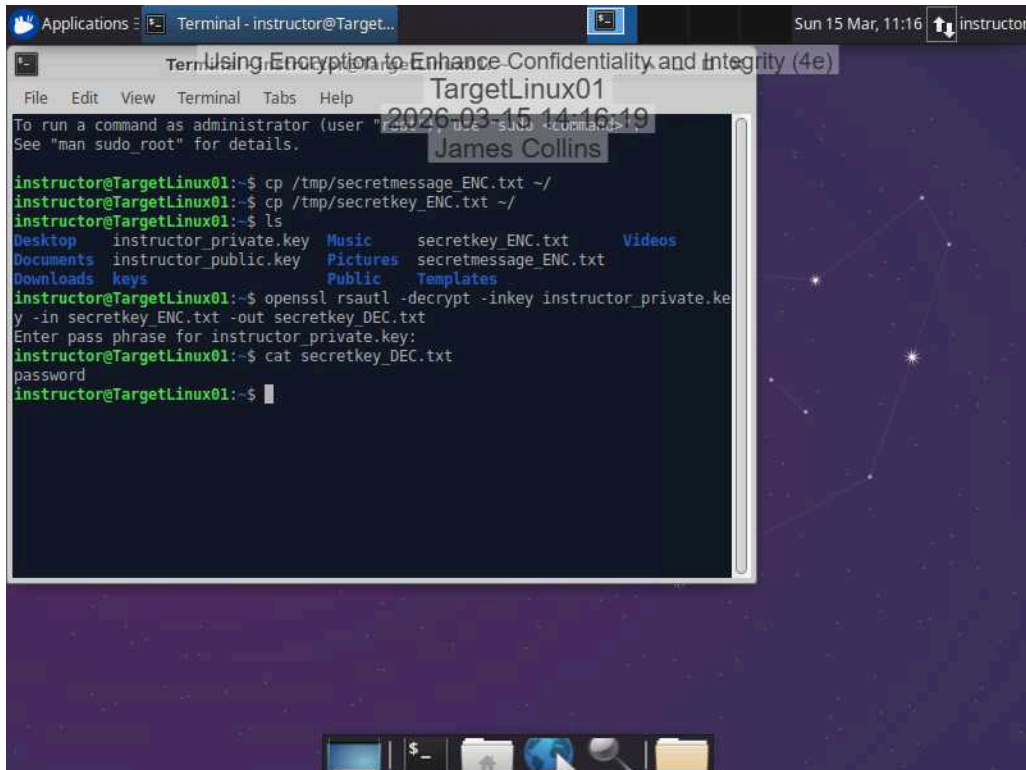


```
student@TargetLinux01:~$ cat secretkey.txt
the password you used to encrypt the secretmessage ENCRYPTED.txt to is password
student@TargetLinux01:~$ openssl rsautl -encrypt -inkey /usr/share/ssl/certs/secretkey_public
.key -in secretkey.txt -out secretkey_ENCRYPTED.txt
student@TargetLinux01:~$ cat secretkey_ENCRYPTED.txt
B00<0c>000 0(wW000080s|y>fE b0Jp0-H|0005;080-zqy+0000 000b00Zm00e01Rs00m}b.H0000
06000b3u0;000 0W?S_H00gc:00060000S00l 000v|0k000i00sm0UX00 0000Q000Q$06;06000000
0
0"b000000-00c0000q0000@h000p 0000%- .00A0G000000b0P0.00k0
0-000W000000/00x 0000 0000h000=m0.0sRS0GLCt000)0W<Y$<a0F0c000#00000p$000%00a000
I"l0V00Up00bNu0000g90^S000T@00-0+03.6700R0x/0Z00{W00`L000,hv(00x2006L007{4000+00
0k00000X007)U0StL0000j)0n0-(00400#0t000S00tH|W)0y"00300r300s0j3e0 !0
0%a600
S06S0I000a00]n00-Bm0Bn40000R0T_a-0RI0[D0=0y0LZ
09fg900000000/00i0000b0m0yS00000F0c00000Sh#000
70"000b0lw050.00(0c000J03f000
v0D0j0b0J000J0
0E00J0a-0H,04F0"n00
000*0{0Y00
0G00P80Y00[00 000<4zZU200}900800j02^c000;j0000001000K{d(0c0b000_08000!a08LLs170
000zc00000700009m0h0_@!r0z000h000=T0u00ZF0,
0-f0)750,00000M000I0Se<e000mM000000L3w0B2%Zy0080'student@TargetLinux01:~$
```

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17. Make a screen capture showing the **decrypted contents of the secretkey_DECRYPTED.txt file.**



The screenshot shows a terminal window titled "Terminal - instructor@Target..." on a desktop environment. The terminal displays the following commands and output:

```
instructor@TargetLinux01:~$ cp /tmp/secretmessage_ENC.txt ~/
instructor@TargetLinux01:~$ cp /tmp/secretkey_ENC.txt ~/
instructor@TargetLinux01:~$ ls
Desktop  instructor_private.key  Music      secretkey_ENC.txt  Videos
Documents instructor_public.key   Pictures   secretmessage_ENC.txt
Downloads keys                   Public     Templates

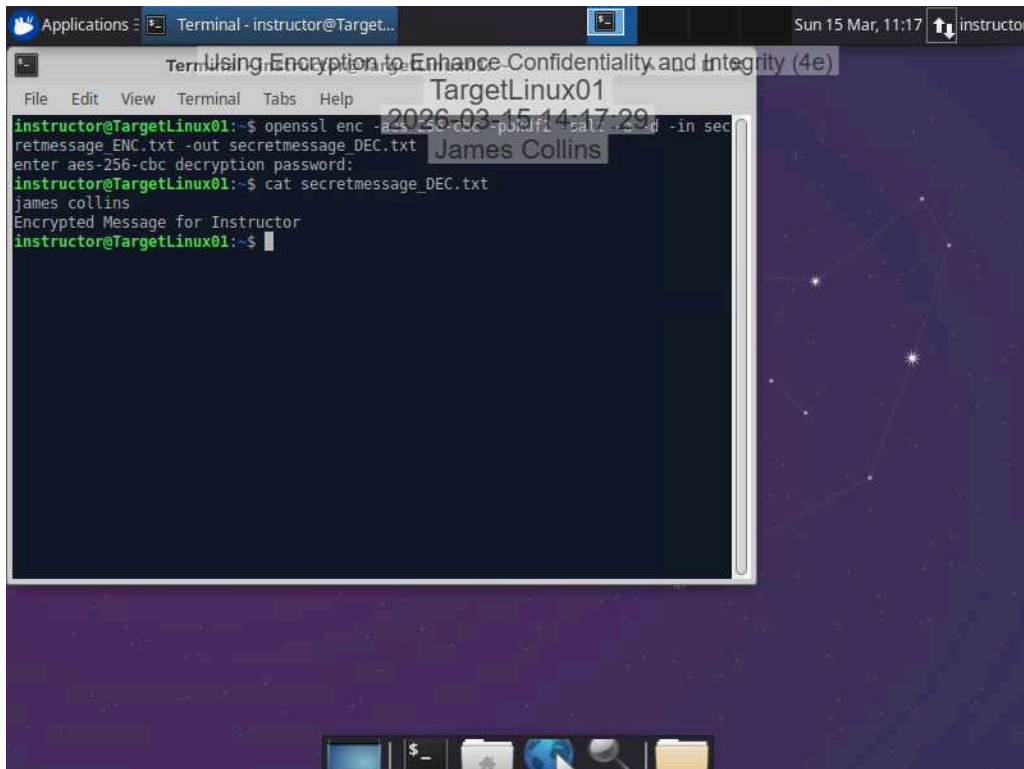
instructor@TargetLinux01:~$ openssl rsautl -decrypt -inkey instructor_private.ke
y -in secretkey_ENC.txt -out secretkey_DEC.txt
Enter pass phrase for instructor_private.key:
instructor@TargetLinux01:~$ cat secretkey_DEC.txt
password
instructor@TargetLinux01:~$
```

Overlaid on the terminal window is a semi-transparent box containing the text: "Using Encryption to Enhance Confidentiality and Integrity (4e)", "TargetLinux01", "2026-03-15 14:16:19", and "James Collins". The desktop background is a dark purple space-themed wallpaper with a constellation. The system tray at the bottom shows icons for a terminal, file manager, and other applications.

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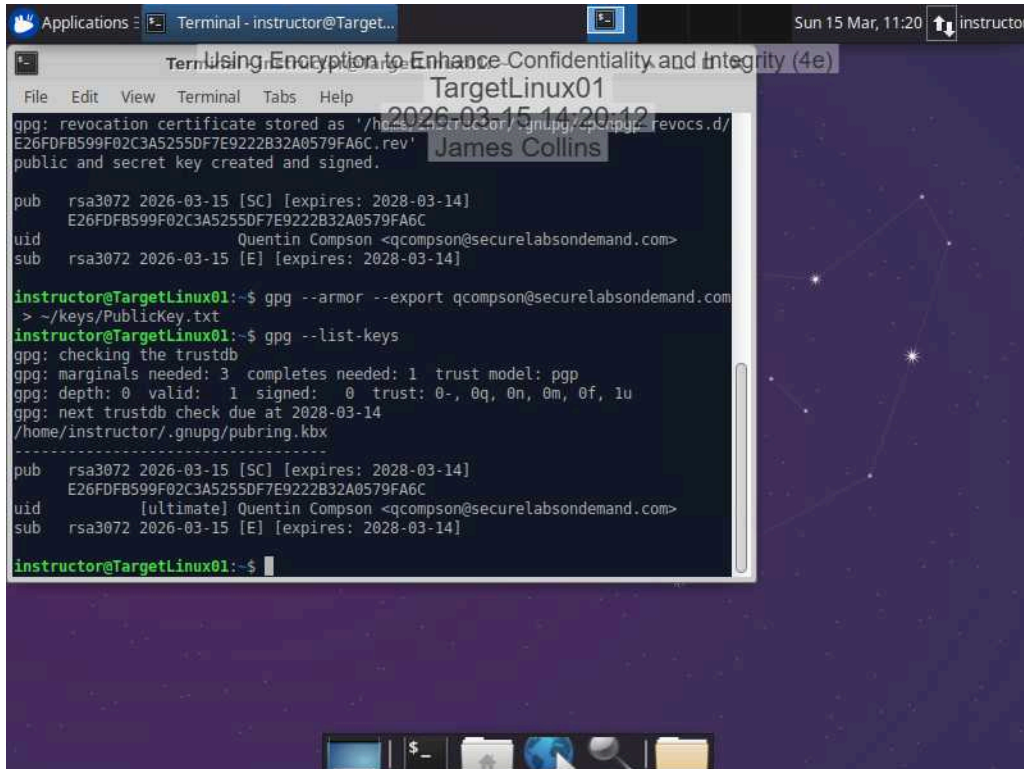
21. Make a screen capture showing the contents of the `secretmessage_DECRYPTED` file.



Section 3: Challenge and Analysis

Part 1: Digitally Sign a Document Using GPG

Make a screen capture showing the **key fingerprint** for the key pair you generated in this part of the lab.



```
Applications Terminal - instructor@TargetLinux01 Sun 15 Mar, 11:20 instructor
Using Encryption to Enhance Confidentiality and Integrity (4e)
TargetLinux01
2026-03-15 14:20:12
James Collins
gpg: revocation certificate stored as '/home/instructor/.gnupg/revocs.d/
E26FDFB599F02C3A5255DF7E9222B32A0579FA6C.rev'
public and secret key created and signed.

pub  rsa3072 2026-03-15 [SC] [expires: 2028-03-14]
     E26FDFB599F02C3A5255DF7E9222B32A0579FA6C
uid          Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-03-15 [E] [expires: 2028-03-14]

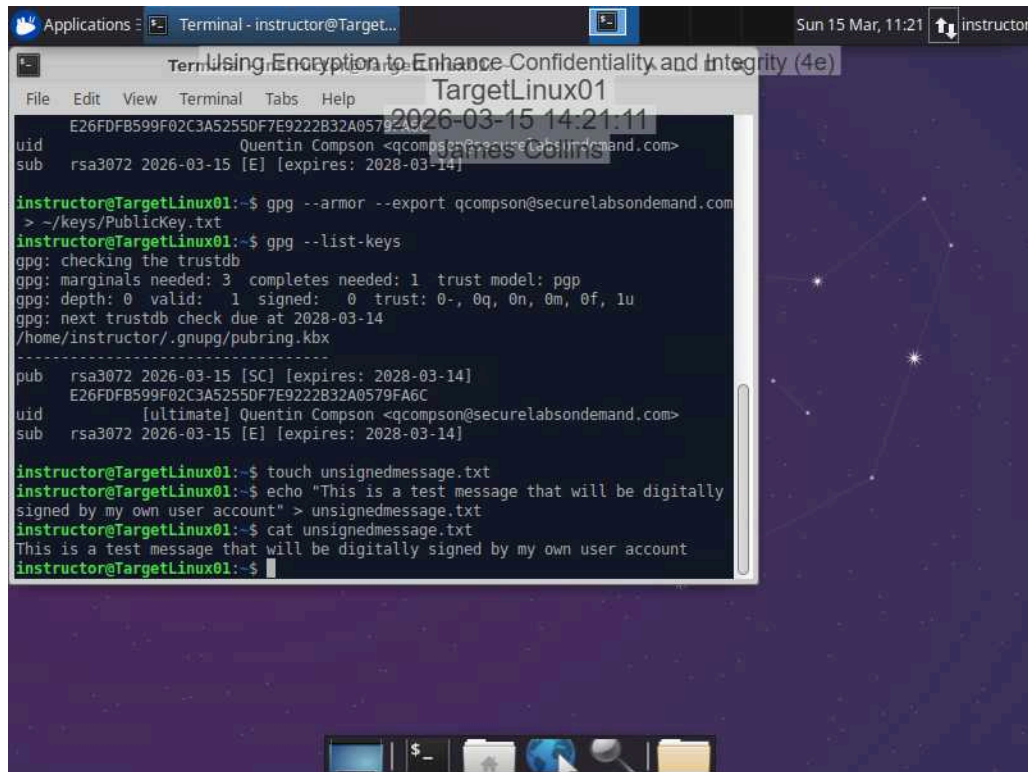
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com
> ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-03-14
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-03-15 [SC] [expires: 2028-03-14]
     E26FDFB599F02C3A5255DF7E9222B32A0579FA6C
uid          [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-03-15 [E] [expires: 2028-03-14]

instructor@TargetLinux01:~$
```

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Make a screen capture showing the contents of the unsignedmessage.txt file.



The screenshot shows a terminal window titled 'Terminal - instructor@TargetLinux01' with a dark background and light text. The terminal output shows the following commands and results:

```
instructor@TargetLinux01:~$ gpg --armor --export qcompson@securelabsondemand.com > ~/keys/PublicKey.txt
instructor@TargetLinux01:~$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-03-14
/home/instructor/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-03-15 [SC] [expires: 2028-03-14]
     E26FDFB599F02C3A5255DF7E9222B32A0579FA6C
uid  [ultimate] Quentin Compson <qcompson@securelabsondemand.com>
sub  rsa3072 2026-03-15 [E] [expires: 2028-03-14]

instructor@TargetLinux01:~$ touch unsignedmessage.txt
instructor@TargetLinux01:~$ echo "This is a test message that will be digitally signed by my own user account" > unsignedmessage.txt
instructor@TargetLinux01:~$ cat unsignedmessage.txt
This is a test message that will be digitally signed by my own user account
instructor@TargetLinux01:~$
```

The terminal window is overlaid on a desktop environment with a dark purple background featuring a constellation of stars. The top of the window shows the system clock as 'Sun 15 Mar, 11:21' and the user 'instructor'.

Part 2: Verify the Digital Signature Using Kleopatra

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Make a screen capture showing the successful signature verification on the signed message file.

